2 Vector

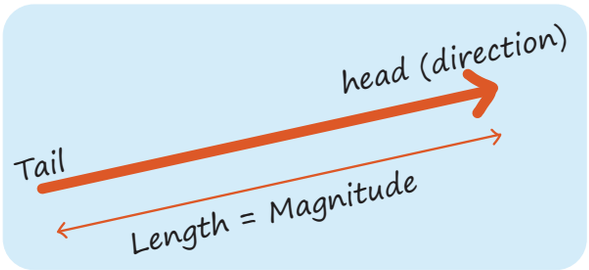
1 Vectors

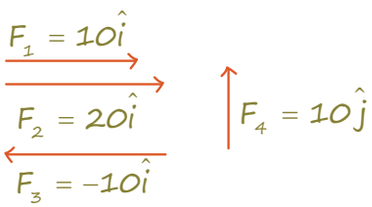
Bina direction ke vector ka information, pura nahi hota. Vector aate hi direction jaroor dekhna.

Vector direction ki topi pahan kar chalta hai.

Scalar Quantity	Vector Quantity
✦ Having Magnitude only	✦ Having Magnitude and direction
✦ Follow simple algebric addition	✦ follow triangle law of vector addition.
✦ Can be changed only by changing its value	✦ Can be changed by changing magnitude only, or changing direction only or changing both.
Ex-Speed, time, Mass, Volume, density current. energy, pressure, flux, electromotive force (volt) etc.	Ex-Force, Velocity, current density, torque, weight etc.

Graphical representation of vector





Mathematical representation

$$\vec{A} = A\hat{A}$$ (Unit vector)

$\hat{A}$: direction of $\vec{A}$

A: Magnitude of $\vec{A}$

$$\text{(direction) } \hat{A} = \frac{\vec{A}}{A}$$

$\hat{i}$, $\hat{j}$ & $\hat{k}$ are unit vector (direction) along x, y and z-axis.

We may assume $\hat{i}$ (east), $-\hat{i}$(west), $\hat{j}$(North) – $\hat{j}$ (south),

$\hat{k}$(up ward), $-\hat{k}$ (downward)

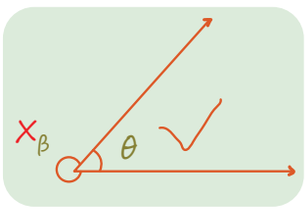
Q1. Force $\vec{F} = 40N\hat{j}$ then find value of $\hat{F}$.

Ans. 1m in North.

Properties of Vectors

- In a vector +ve sign and sign –ve indicate direction only.

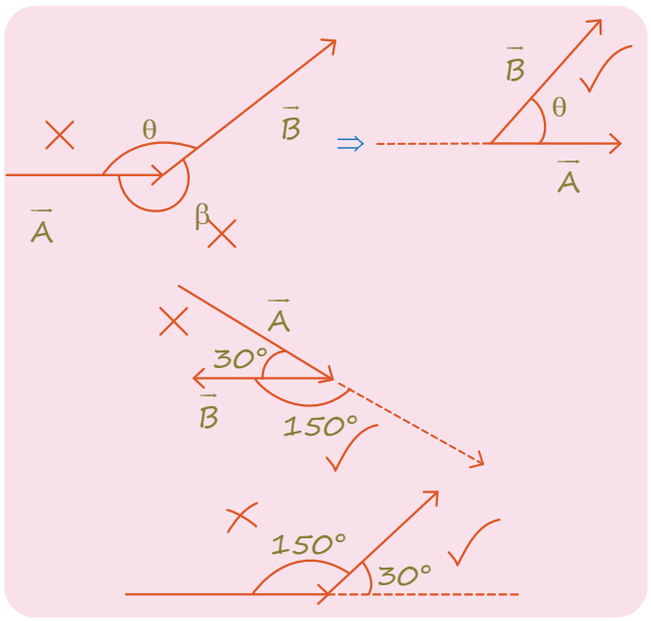
  Ex: +5N and –5N, same magnitude of force in opposite direction.
- Angle between vector – When two vectors are placed head to head or tail to tail then smaller angle between vector is called angle between vector.



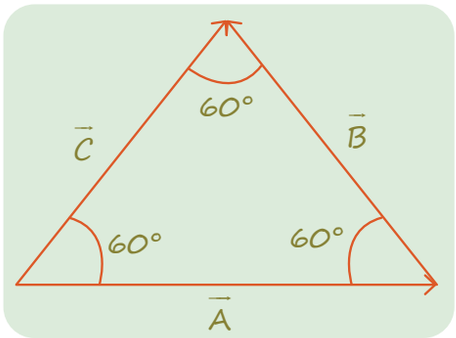
- Vector can be shifted parallel to itself by keeping magnitude and direction fixed.
- Rotation of vector not allowed it will change meaning of vector (Angle other than 360°).
- If angle between $\vec{A}$ and $\vec{B}$ vector is θ then angle between $\vec{A}$ and $-\vec{B}$ is (180°-θ).
- All equal vectors are parallel but all parallels are not equal.

✦ All opposite (Negative) Vectors are Antiparallel but all antiparallel are not Opposite Vector

Angle Between Vectors:



Q.2. Find Angle b/w vector $\vec{A}$, $\vec{B}$ & $\vec{C}$



Sol. Angle b/w $\vec{A}$ & $\vec{B}$ = 120°

$\vec{B}$ & $\vec{C}$ = 60°

$\vec{A}$ & $\vec{C}$ = 60°

✦ When vector multiplied with positive scalar → direction remains same only magnitude will change. If –ve scalar then direction reversed & magnitude change.

Q.3. If $\vec{A} = 30m\,\hat{i}$, then:

(i) $3\vec{A} = 90m\,\hat{i}$

(ii) $\frac{\vec{A}}{3} = \frac{30m}{3}\hat{i} = 10m\,\hat{i}$

(iii) $-5\vec{A} = -5(30m\,\hat{i}) = -150m\,\hat{i}$

(iv) $|\vec{A}| = 30m$

(v) $\hat{A} = \hat{i}$

Q.4. Some forces are given then draw

(1) $\vec{F}_1$ = 10N along east

(2) $\vec{F}_2$ = 20 N along north

(3) $\vec{F}_3$ = 40 N west

(4) $\vec{F}_4$ = 50 N North west

(5) $\vec{F}_5$ = 40 N South-east

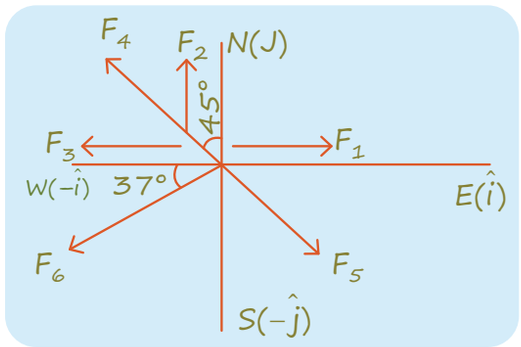
(6) $\vec{F}_6$ = 10 N at 37° South-west

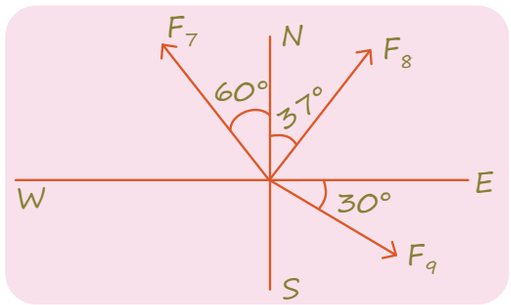
(7) $\vec{F}_6$ = 10 N at 60° west of North

(8) $\vec{F}_8$ = 10 N at 37° east of North

(9) $\vec{F}_9$ = 8N, east 30° South

Sol.





Q.5. If angle b/w vector $\vec{A}$ and $\vec{B}$ is 30° then angle b/w

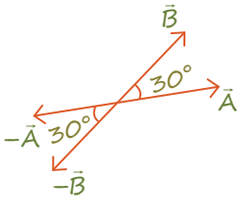
Sol. (a) $\vec{A}$ & $-\vec{B}$ = 150°

(b) $-\vec{A}$ & $\vec{B}$ = 150°

(c) $-\vec{A}$ & $-\vec{B}$ = 30°

(d) $\vec{A}$ & $2\vec{B}$ = 30°

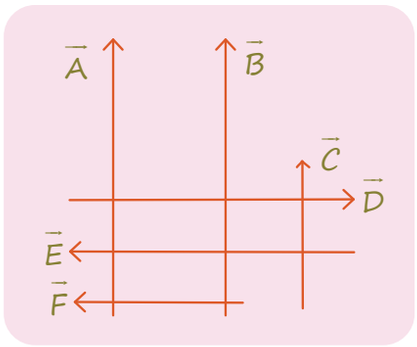
(e) $2\vec{A}$ & $-3\vec{B}$ = 150°



$2\vec{A}$ & $\vec{A}$ ka direction same hi hoga lala.

2 Type of Vectors

S. No.	Type	Magnitude	Direction\Angle
1.	Equal Vector	Same	Same ($\theta = 0$)
2.	Parallel Vector	May or May not same	Same ($\theta = 0$)
3.	Opposite Vector or Negative Vectors	Same	Opposite $\theta = 180°$
4.	Anti-parallel Vector	May or May not same	$\theta = 180°$ opposite
5.	Orthogonal	May same	$\theta = 90°$
6.	Zero/Null Vector	Zero	any direction
7.	Unit Vectors	One	$\hat{A} = \frac{\vec{A}}{A}$
8.	Co-linear vector	may or may be same	Angle 0° or 180°
9.	Concurrent vector	Point of Action same	magnitude angle kuch bhi ho skta hai
10.	Coplanar vector	Vectors are in same plane	
11.	Axial vectors	Associated with rotational motion	Perpendicular to the plane of motion



Parallel Vector	$(\vec{A},\vec{B})$ $(\vec{B},\vec{C})$ $(\vec{C},\vec{A})$ $(\vec{E},\vec{F})$
Equal vector	$(\vec{A},\vec{B})$
Anti-Parallel	(D, E) & (D, F)
Opposite or Negative	(D, E)
Orthogonal	$\vec{A}$ with D, E & F $\vec{B}$ with D, E & F $\vec{C}$ with D, E & F

NOTE: $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and

$\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ and

$\frac{A_x}{B_x} = \frac{A_y}{B_y} = \frac{A_z}{B_z} = n$. If n – +ve then $\vec{A} \parallel_r \vec{B}$

If n = –ve then $\vec{A}$ anti $\parallel_r$ to $\vec{B}$

Ex: $\vec{A} = 2i - 4\hat{j} + 3k$

$\vec{B} = 6i - 12j + 9k$

} $\vec{A}$ & $\vec{B}$ are $\parallel_r$

ii. $\vec{A} = 2i - 4j - 3\hat{k}$

$\vec{B} = -6\hat{i} + 12\hat{j} + 9\hat{k}$

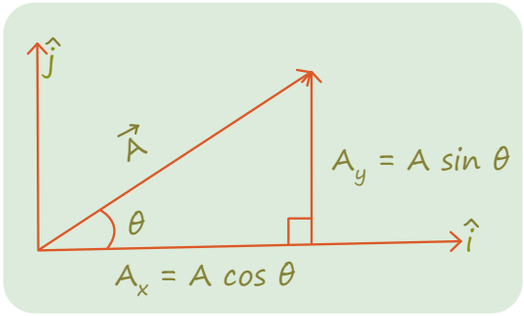
} $\vec{A}$ & $\vec{B}$ are anti $\parallel r$

iii. $\vec{A} = 2\hat{i} - 3\hat{j} + 4\hat{k}$

$\vec{B} = 8i + 12j - 16\hat{k}$

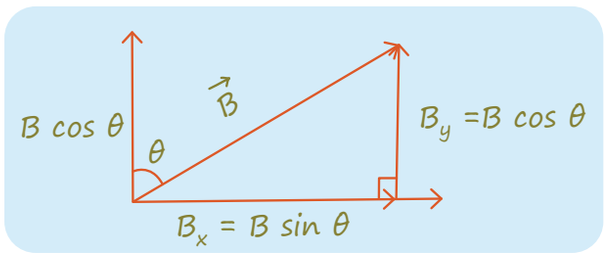
} Not $\parallel r$ or Anti $-\parallel r$

Components of Vector in 2-D (effect of Vector)



- $\vec{A} = A_x\hat{i} + A_y\hat{j}$

  $\vec{A} = A\cos\theta\ \hat{i} + A\sin\theta\ \hat{j}$



- $\vec{B} = B_x\hat{i} + B_y\hat{j} = B\sin\theta\,\hat{i} + B\cos\theta\,\hat{j}$

Magnitude of resultant Vector

$A = \sqrt{A_x^2 + A_y^2}$ or $B = \sqrt{B_x^2 + B_y^2}$

Direction:

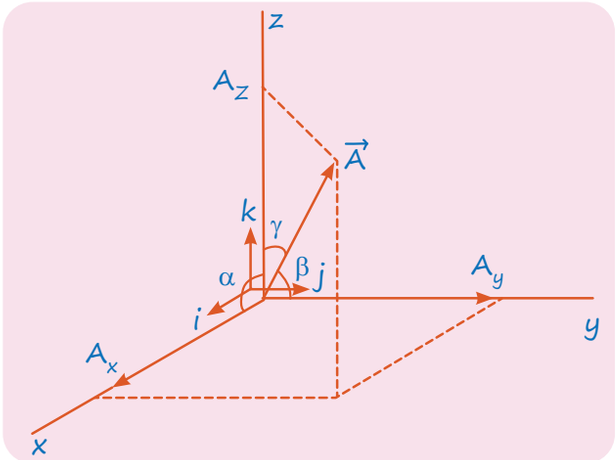
$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$ or $\theta = \tan^{-1}\left(\frac{B_x}{B_y}\right)$

Rectangular component of a vector in 3D

$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$

Magnitude

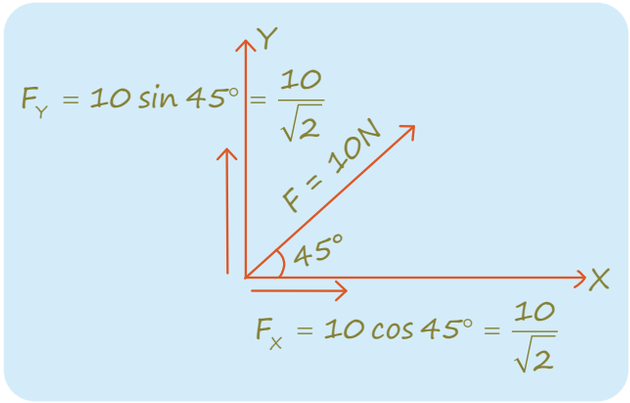
$|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$

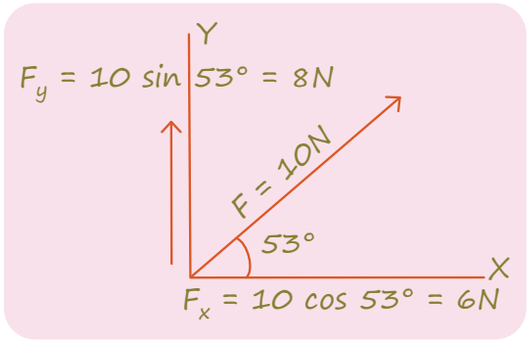


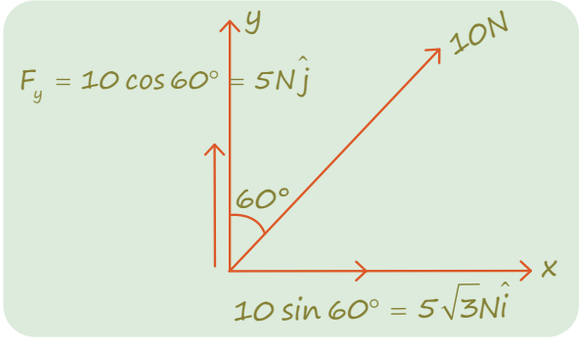
- If vector is making an angle α, β and γ from x, y and z-axis respectively then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$; $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

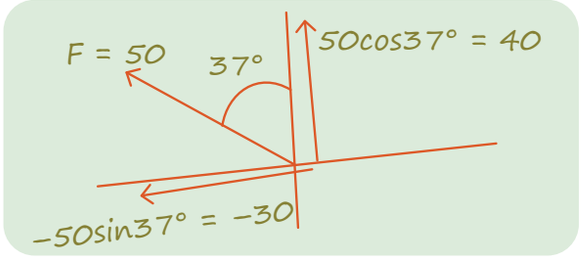
Direction Cosine

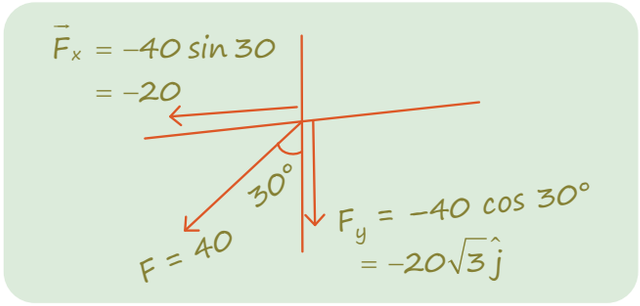
$\cos\alpha = \frac{A_x}{A}$ $\quad \cos\beta = \frac{A_y}{A}$ $\quad \cos\gamma = \frac{A_z}{A}$

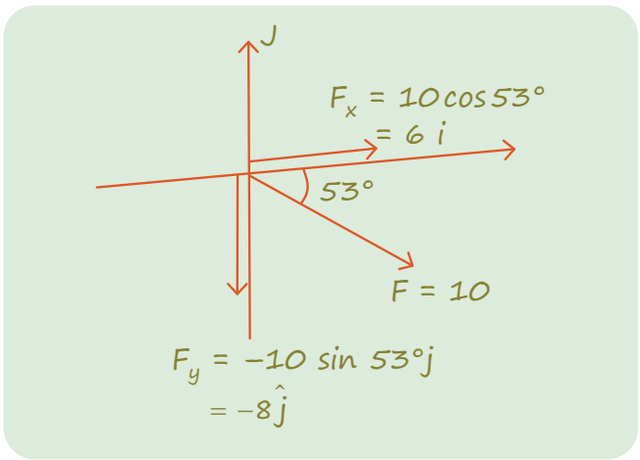












Q.6. A vector $2\hat{i}$ rotated, then new vector will be

Sol. New Vector

(i) By $\theta = 90° \rightarrow 2\hat{j}$

(ii) By $\theta = 180° \rightarrow -2\hat{i}$

(iii) By $\theta = 45° \rightarrow \sqrt{2}\hat{i} + \sqrt{2}\hat{j}$

(iv) By $\theta = 270° \rightarrow -2\hat{j}$

(v) By $\theta = 360° \rightarrow +2\hat{i}$

MR* Box

In rotation of vector magnitude remains same and direction changes.

Q.7. If vector $2\hat{i} + 2\hat{j}$ rotated then new vector will be

Sol. New Vector

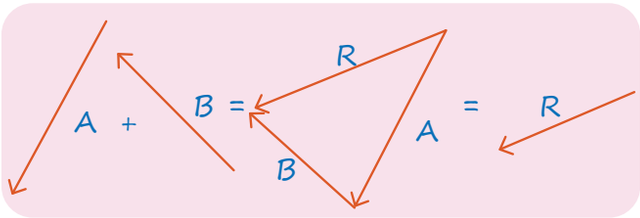
(i) $\theta = 45° \rightarrow 2\sqrt{2}\hat{j}$

(ii) $\theta = 90° \rightarrow -2\hat{i} + 2\hat{j}$

(iii) $\theta = 180° \rightarrow -2\hat{i} - 2\hat{j}$

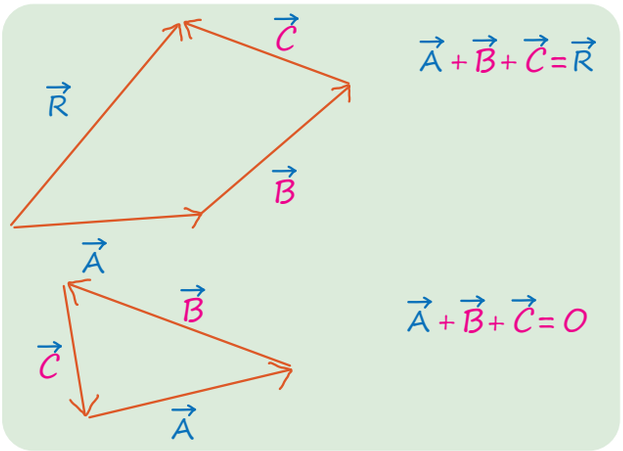
3 Vectors addition

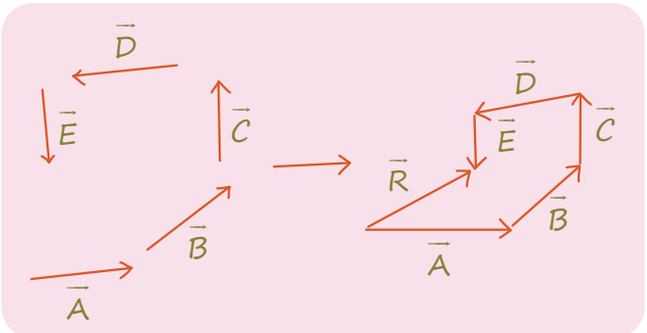
- It is the process of combining two vectors by placing the tail of one vector at the head of the other. i.e.,



Polygon Law of vector addition

- Start tail of next vector from head of previous vector and so on.





Find Magnitude of Vector

(i) $\vec{A} = 2\hat{i} + 3\hat{J}$

$|\vec{A}| = \sqrt{2^2 + 3^2} = \sqrt{13}$

(ii) $\vec{B} = 3\hat{i} + 4\hat{J}$

$|\vec{B}| = \sqrt{3^2 + 4^2} = 5$

(iii) $\vec{C} = 3\hat{i} + 4\hat{J} + 5\hat{k}$

(iv) $|\vec{C}| = \sqrt{3^2 + 4^2 + 5^2} = \sqrt{50}$

$\vec{D} = \hat{i} - \hat{j} + \hat{k}$

(v) $|\vec{D}| = \sqrt{1^2 + (-1)^2 + (1)^2} = \sqrt{3}$

(iv) $\vec{E} = 6i - 8\hat{j} + 10\hat{k}$

$|\vec{E}| = \sqrt{36 + 64 + 100} = \sqrt{200} = 10\sqrt{2}$

(vii) $\vec{F} = 10\hat{i} - 10\hat{j} - 10\hat{k}$

$|\vec{F}| = \sqrt{100 + 100 + 100} = 10\sqrt{3}$

Q.8. Vector $\vec{A} = 5\hat{i} + 5\sqrt{3}\hat{j}$ then find angle of vector $\vec{A}$ from x-axis

Sol. $|\vec{A}| = \sqrt{5^2 + (5\sqrt{3})^2} = 10$

Method-1

$\cos\alpha = \frac{5}{10} = \frac{1}{2} \Rightarrow \alpha = 60$

Method-2

$\tan\alpha = \frac{5\sqrt{3}}{5} \Rightarrow \tan\alpha = \sqrt{3}$

$\alpha = 60°$

Q.9. The angles which a vector $\hat{i} + \hat{j} + \sqrt{2}\,\hat{k}$ makes with X, Y and Z axes respectively are

Sol. $|\vec{A}| = \sqrt{1^2 + 1^2 (\sqrt{2})^2} = \sqrt{1 + 1 + 2} = 2$

$\cos\alpha = \frac{1}{2} \Rightarrow \alpha = 60°, \beta = 60°$

$\cos\gamma = \frac{A_z}{A} = \frac{\sqrt{2}}{2} \Rightarrow \cos\gamma = \frac{1}{\sqrt{2}} \Rightarrow \gamma = 45°$

Q.10. Find unit vector $\hat{A} = \frac{\vec{A}}{A}$

(1) $\vec{A} = \hat{i}$ $\quad \hat{A} = \hat{i}$

(2) $\vec{B} = \hat{i} + \hat{j}$ $\quad \hat{B} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$

(3) $\vec{C} = \hat{i} + \hat{j} - \hat{k}$ $\quad \hat{C} = \frac{\hat{i} + \hat{j} - \hat{k}}{\sqrt{3}}$

(4) $\vec{D} = \sin\theta\,\hat{i} - \cos\theta\,\hat{j}$ $\quad \hat{D} = \sin\theta\hat{i} - \cos\theta\hat{j}$

(5) $\vec{E} = 3\hat{i} + 4\hat{j}$ $\quad \hat{E} = \frac{3\hat{i} + 4\hat{j}}{5}$

MR* Box

Vector addition

$\vec{R} = \vec{A} + \vec{B}$ = vector addition of $\vec{A}$ & $\vec{B}$ = Resultant of $\vec{A}$ & $\vec{B}$

$|\vec{A}| + |\vec{B}|$ = sum of magnitude of $\vec{A}$ & $\vec{B}$

(direction nahi lena, pahle magnitude nikaalo then direct add karo)

$|\vec{R}| = |\vec{A} + \vec{B}|$ = magnitude of resultant of $\vec{A}$ & $\vec{B}$ = magnitude of resultant (pahle direction ke sath add karo, then magnitude nikaalo)

$\vec{D} = \vec{A} - \vec{B}$ = vector subtraction of $\vec{A}$ & $\vec{B}$

$|\vec{D}| = |\vec{A} - \vec{B}|$ = magnitude of vector subtraction

$|\vec{A}| - |\vec{B}|$ = subtraction of magnitude of vector

Vector addition / subtraction in terms of component:

$\vec{A} = A_x\,\hat{i} + A_y\,\hat{j} + A_z\hat{k}$, $\vec{B} = B_x\,\hat{i} + B_y\,\hat{j} + B_z\,\hat{k}$

$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$

$\vec{A} - \vec{B} = (A_x - B_x)\hat{i} + (A_y - B_y)\hat{j} + (A_z - B_z)\hat{k}$

$\vec{B} - \vec{A} = (B_x - A_x)\hat{i} + (B_y - A_y)\hat{j} + (B_z - A_z)\hat{k}$

Q.11. $\vec{A} = 4\hat{i} + 3\hat{j}$; $\vec{B} = 3\hat{i} - 4\hat{j}$ then find

Sol. $\vec{A} + \vec{B} = 7\hat{i} - \hat{j}$

$\vec{A} - \vec{B} = \hat{i} + 7\hat{j}$

$\vec{B} - \vec{A} = -\hat{i} - 7\hat{j}$

$|\vec{A}| + |\vec{B}| = 5 + 5 = 10$

$|\vec{A} + \vec{B}| = \sqrt{(49 + 1)} = \sqrt{50}$

$|\vec{A} - \vec{B}| = \sqrt{50}$

Q.12. Two vectors $\vec{A} = 6\hat{i}$ & $\vec{B} = 8\hat{j}$ then

Sol. (i) $\vec{A} + \vec{B} = 6\hat{i} + 8\hat{j}$

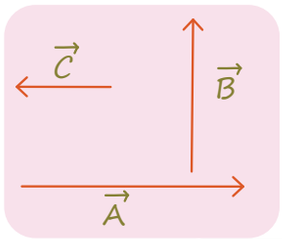
(ii) $\vec{A} - \vec{B} = 6\hat{i} - 8\hat{j}$

(iii) $|\vec{A}| + |\vec{B}| = 14$

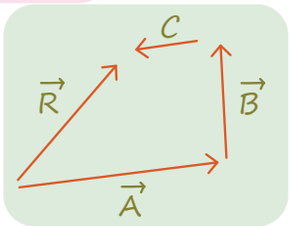
(iv) $|\vec{A} + \vec{B}| = 10$

MR* Box

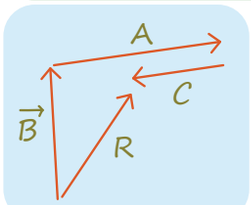
Angle nikalne ke liye tail–tail or head–head ko ek sath rakho, vector add karne ke liye 1st ke head se 2nd ka tail start karo. ek bat or samajh lo jin 2–3 vector ko add kar rahe ho unme draw karne ka order kuch bhi le sakta ho, niche wale diagram se samjho. In all 3-diagram $\vec{R}$ is same.



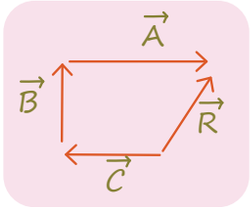
$\vec{R} = \vec{A} + \vec{B} + \vec{C} \Rightarrow$



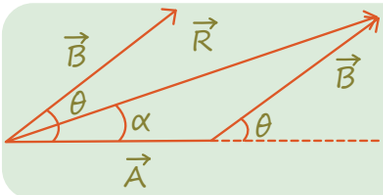
$\vec{R} = \vec{B} + \vec{A} + \vec{C} \Rightarrow$



$\vec{R} = \vec{C} + \vec{B} + \vec{A} \Rightarrow$



Triangle Law of Vector addition



resultant must be in the plane of $\vec{A}$ and $\vec{B}$.

- $|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$
- $\tan\alpha = \dfrac{B\sin\theta}{A + B\cos\theta}$

If $\theta = 0°$	$\theta = 90°$	$\theta = 180°$
$R_{max} = A + B$	$R = \sqrt{A^2 + B^2}$	$R_{min} = A - B$

- $A - B \le R \le A + B$

If $|\vec{A}| = |\vec{B}| = A$ and Angle b/w them θ

$|\vec{R}| = 2A\cos(\theta/2)$ $\quad$ $|\vec{D}| = 2A\sin(\theta/2)$

Vector Subtraction

Angle B/w $\vec{A}$ & $\vec{B}$ is θ then $\vec{D} = \vec{A} - \vec{B}$

Magnitude

$|\vec{D}| = \sqrt{A^2 + B^2 - 2AB\cos\theta}$

If $\theta = 0°$	$\theta = 90°$	$\theta = 180°$
$D_{min} = A - B$	$D = \sqrt{A^2 + B^2}$	$D = A + B$

- $A - B \le D \le A + B$

4 Two vector of same magnitude

$\theta = 0°$	$\theta = 60°$	$\theta = 90°$	$\theta = 120°$	$\theta = 180°$
$R = 2A$	$R = \sqrt{3}A$	$R = \sqrt{2}A$	$R = A$	$R = 0$
$D = 0$	$D = A$	$D = \sqrt{2}A$	$D = \sqrt{3}A$	$D = 2A$

Properties of vector addition and substraction:

- Magnitude of Vector addition and subtraction are same at 90°.
- $\vec{A} + \vec{B} = \vec{B} + \vec{A}$ Commutative
- $n(\vec{A} + \vec{B}) = n\vec{A} + n\vec{B}$ distributive
- $\vec{A} - \vec{B} \ne \vec{B} - \vec{A}$
- $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$ and $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$ then $\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$

Q.13. Which of the following combinations of three forces can give zero resultant?

1) [2, 4, 7] 2) [3, 1, 5]
3) [2, 8, 11] 4) [3, 4, 2]

Sol. For three forces to give zero resultant, the third force must be equal to the resultant of the other two forces.

The resultant of two forces A and B lies in the range:

$|A - B| \le C \le (A + B)$

Checking options:

1) [2, 4, 7]

$4 + 2 = 6 < 7 \rightarrow$ Not possible

2) [3, 1, 5]

$3 + 1 = 4 < 5 \rightarrow$ Not possible

3) [2, 8, 11]

$8 + 2 = 10 < 11 \rightarrow$ Not possible

4) [3, 4, 2]

$|4 - 3| = 1$ and $4 + 3 = 7$

2 lies between 1 and 7 $\rightarrow$ Possible

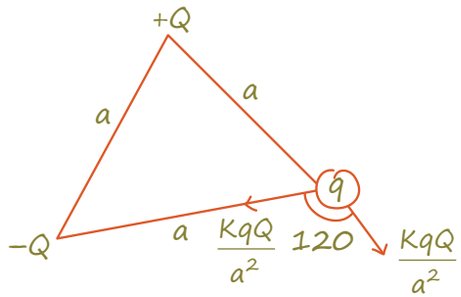
Final Answer:

Correct combination is [3, 4, 2]

MR* Box

Koi bhi 2-vector lo uske resultant ke range (A − B to A + B) me 3rd vectors aaya to sum of all three vector zero ho skta hai.

Q.14. Find electrostatic force on q charge



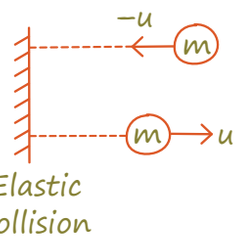
$F_{net} = \frac{kqQ}{a^2}$

If both Q charge of same nature then

$F_{net} = \frac{\sqrt{3}kQq}{a^2}$

Question on Vector Subtraction

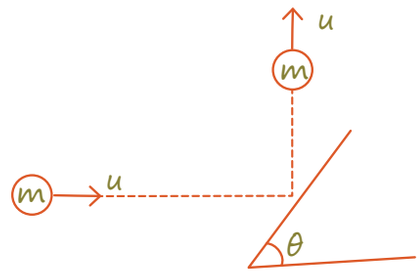
Q.15.



Elastic collision

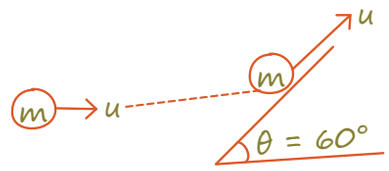
Sol. $\Delta p = m(\vec{v}_f - \vec{v}_i) = 2mu$

Q.16.



Sol. $\Delta p = m(\vec{v}_f - \vec{v}_i) = m\sqrt{2}\,u$

Q.17.



Sol. $\Delta p = m(\vec{v}_f - \vec{v}_i) = mu$

Angle between initial and final velocity is 60°

Q.18. If the sum of two unit vectors is a unit vector, then the magnitude of their difference is:

Sol. $|\vec{A}| = |\vec{B}| = 1$, $|\vec{R}| = 1$ (given)

Angle between $\vec{A}$ and $\vec{B}$ = 120°

Vector subtraction of two unit vectors = $\sqrt{3}$

Q.19. Initial velocity v = 10 $\hat{i}$ East After some time its velocity becomes 10 m/s North, then Find change in velocity is

Sol. $\vec{v}_i = 10\hat{i}$

$\vec{v}_f = 10\hat{j}$

$\Delta v = \vec{v}_f - \vec{v}_i = 10\hat{j} - 10\hat{i}$

$|\Delta v| = 10\sqrt{2}$ (North-West)

Change in magnitude of velocity = Zero

Q.20. If $\vec{A} = 2\hat{i} + 2\hat{j}$ and $\vec{B} = 3\hat{i} + 3\hat{k}$, then find:

Sol. (i) $\vec{A}/\vec{B}$ = Not possible

(ii) $|\vec{A}|/\vec{B}$ = Not possible

(iii) $\vec{A}/|\vec{B}| = (2\hat{i} + 2\hat{j})/(3\sqrt{2})$

(iv) $|\vec{A}|/|\vec{B}| = 2\sqrt{2}/3\sqrt{2} = 2/3$

Q.21. For two vectors $\vec{A}$ and $\vec{B}$: Vector addition $\vec{R} = \vec{A} + \vec{B}$ Vector subtraction $D = \vec{A} - \vec{B}$ Then angle between $\vec{A}$ and $\vec{B}$ for given conditions:

Sol.

1) $|\vec{A} + \vec{B}| = |\vec{R}| = |\vec{A}| = |\vec{B}| \rightarrow 120°$
2) $|\vec{A}| = |\vec{B}| = |\vec{A} - \vec{B}| \rightarrow 60°$
3) $A^2 + B^2 = |\vec{A} - \vec{B}|^2 = D^2 \rightarrow 90°$
4) $A^2 + B^2 = R^2 \rightarrow 90°$
5) $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}| \rightarrow 90°$
6) $|\vec{A} + B| = \vec{R} = |\vec{A}| + |\vec{B}| \rightarrow 0°$
7) $D = |\vec{A} - \vec{B}| = |\vec{A}| + |\vec{B}| \rightarrow 180°$
8) $|\vec{A} + \vec{B}| = \vec{R} = |\vec{A}| - |\vec{B}| \rightarrow 180°$
9) $|\vec{A} - \vec{B}| = \vec{D} = |\vec{A}| - |\vec{B}| \rightarrow 0°$
10) $|\vec{A} + \vec{B}| = \sqrt{3}\,A = \sqrt{3}\,B \rightarrow 60°$
11) $|\vec{A} - \vec{B}| = \sqrt{3}\,A = \sqrt{3}\,B \rightarrow 120°$

Q.22. Two vector of magnitude 3 and 4 acting at different angle, then find their resultant

Sol.

(i) $\theta = 0° \rightarrow R_{max} = 3 + 4$

(ii) $\theta = 60° \rightarrow$

$R = \sqrt{(3^2 + 4^2 + 2\times3\times4\cos 60°)} = \sqrt{37}$

(iii) $\theta = 90° \rightarrow$

$R = \sqrt{(3^2 + 4^2 + 2\times3\times4\cos 90°)} = 5$

(iv) $\theta = 120° \rightarrow$

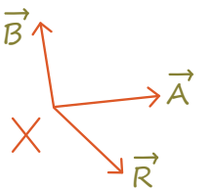
$R = \sqrt{(9 + 16 - 2\times3\times4 \times 1/2)} = \sqrt{13}$

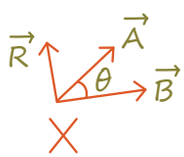
(v) $\theta = 180° \rightarrow R_{min} = 4 - 3 = 1$

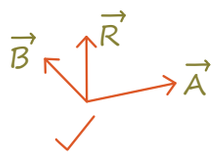
$1 \le R \le 7$

Q.23. Which of the following diagram is correct:

Sol.

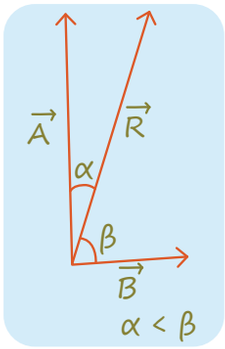
(a) 

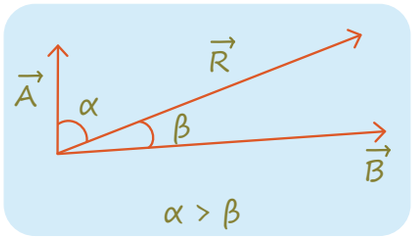
(b) 

(c) 

MR* Box

2-vector ka resultant unke bich me rahega, bade wale vector ke pas niche wale diagram se feel kro.





Q.24. Which pair of two vectors cannot give a resultant of magnitude 4?
(a) 2, 4 (b) 4, 4 (c) 4, 3 (d) 7, 5

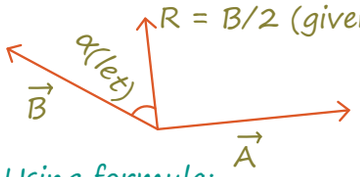
Sol. (c) $R_{min} = 9 - 4 = 5$

$= 9 + 4 = 13$

Resultant 4 nahi hoga.

Q.25. If the resultant of two vectors is half of the bigger vector and is perpendicular to the smaller vector, find the angle between the vectors.

Sol.



Using formula:

$R = B\cos\theta$

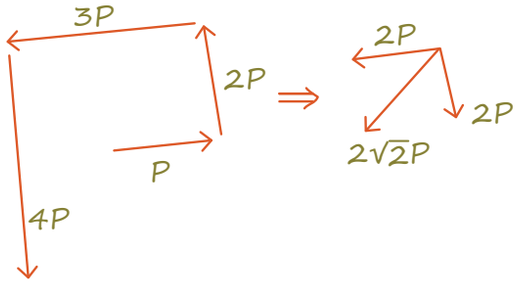
$B/2 = B\cos\theta \Rightarrow \cos\theta = 1/2 \Rightarrow \theta = 60°$

Angle between vectors $\vec{A}$ & $\vec{B}$

$= 60° + 90° = 150°$

Q.26. Four vectors of magnitude p, 2p, 3p, and 4p act along the four sides of a square ABCD in cyclic order. Find the resultant force.

Sol.

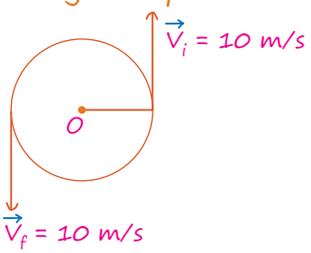


Q.27. Ramlal is moving with velocity 6m/s along east and pinky with 6 m/s at 30° east of north then relative velocity of pinky w.r.t Ramlal.

Sol. $\vec{V}_{PR} = \vec{V}_P - \vec{V}_R$ same vector ka subtraction

at 60° $|\vec{V}_{PR}| = 6$ m/s

Q.28. Find change in Speed and velocity



Sol. Change in speed = 0

Magnitude of change in velocity = 20 m/s

Q.29. If $\vec{A} = 0.6\hat{i} + \beta\hat{j}$ is a unit vector then find value of β.

Sol. $|\vec{A}| = 1$ if A is unit vector

$\sqrt{(0.6)^2 + \beta^2} = 1$

$\beta^2 + 0.36 = 1$

$\beta = \sqrt{0.64} = 0.8$

Q.30. Two force 10N and 6N acting then find resultant of these two force.

Sol. $10 - 6 \le R \le 10 + 6$

R will between 4N to 16N

Q.31. Find the angle which a vector $\hat{i} + \hat{j} + \sqrt{2}\hat{k}$ makes with x, y and z-axis

Sol. $A = \sqrt{1^2 + 1^2 + (\sqrt{2})^2} = 2$

$A_x = 1, A_y = 1, A_z = \sqrt{2}$

$\cos\alpha = \frac{A_x}{A}$ $\cos\beta = \frac{A_y}{A}$ $\cos\gamma = \frac{A_z}{A}$

$\alpha = 60°$ $\beta = 60°$ $\gamma = 45°$

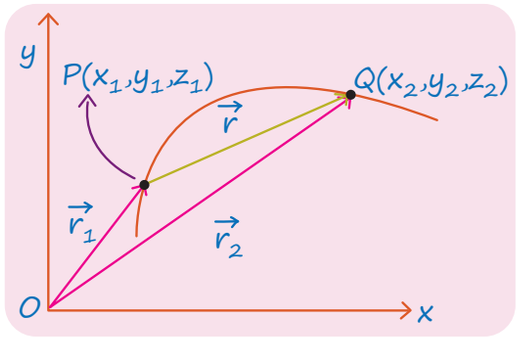
Q.32. In which of the following combination of three force resultant will be zero.

(a) 3N, 7N, 8N (b) 2N, 5N, 1N

(c) 3N, 12N, 7N (d) 4N, 5N, 10N

Sol. (a) Sum of two smaller must be greater or equal to (3rd).

5 Displacement vector



✦ If a particle moves from the initial position (x_1, y_1, z_1) to the final position (x_2, y_2, z_2), the displacement vector r is given by:

$r = (x_2 - x_1)i + (y_2 - y_1)j + (z_2 - z_1)k$

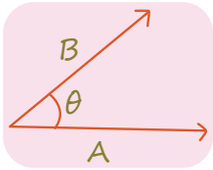
or $r = \Delta xi + \Delta yj + \Delta zk$

Magnitude of displacement vector

✦ $r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

6 Scalar Product (Dot Product)

2-vector ka parallel component chipakna hai, perpendicular component ko ignore karna hai.



✦ $\vec{A}\cdot\vec{B} = A(B\cos\theta) = A$(Component of B along A)

$= (A\cos\theta)\,B = B$(Component of A along B)

Component of B along $A = \frac{\vec{A}\cdot\vec{B}}{A}$

Component of A along $B = \frac{\vec{A}\cdot\vec{B}}{B}$

✦ Result of dot product is always scalar.

$\hat{i}\cdot\hat{i} = 1$ $\hat{j}\cdot\hat{j} = 1$ $\hat{k}\cdot\hat{k} = 1$

$\hat{i}\cdot\hat{k} = 0$ $\hat{j}\cdot\hat{i} = 0$ $\hat{k}\cdot\hat{j} = 0$

\# $\vec{A}\cdot\vec{B} = (A_x\hat{i} + A_y\hat{j} + A_z\hat{k})\cdot(B_x\hat{i} + B_y\hat{j} + B_z\hat{k})$

$= A_xB_x + A_yB_y + A_zB_z$

Application of dot Product

- To Find Angle B/W vectors

$\vec{A}\cdot\vec{B} = AB\cos\theta$

$\cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB}$

- To check unit vector

If $\vec{A}$ is a unit vector then $\vec{A}\cdot\vec{A} = 1$

- To check perpendicular vector (orthogonal)

If $\vec{A}\cdot\vec{B} = AB\cos 90° = 0$

$\vec{A}\cdot\vec{B} = 0 \quad (\vec{A} \perp r \vec{B})$

- To find component of one vector along other.

$\vec{A}\cdot\vec{B} = A(B\cos\theta)$

$B\cos\theta = \frac{\vec{A}\cdot\vec{B}}{A}$ = Comp^n of B along A

Q.33. If $\vec{A} = 4\hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{B} = \hat{i} + 3\hat{j} - 2\hat{k}$ then find $\vec{A}.\vec{B}$

(1) 10 (2) 16 (3) 3 (4) 14

Sol. (2) $\vec{A}.\vec{B} = (4\hat{i} + 2\hat{j} - 3\hat{k})\cdot(\hat{i} + 3\hat{j} - 2\hat{k})$

$= 4 + 6 + 6 = 16$

Q.34. A force $\vec{F} = 4\hat{i} + 5\hat{j}$ newton displaces a particle through $\vec{S} = 3\hat{i} + 6\hat{j}$ meter. Then find work done.

(1) 24 Joule (2) 12 Joule

(3) 28 Joule (4) 42 Joule

Sol. (4) $W = \vec{F}.\vec{S} = (4\hat{i} + 5\hat{j})\cdot(3\hat{i} + 6\hat{j})$

$= 12 + 30$

$= 42$ J

Q.35 If $\vec{A} = \hat{i} + \sqrt{3}\hat{j}$ and $\vec{B} = \hat{i}$ then find angle B/W $\vec{A}$ & $\vec{B}$

Sol. $\vec{A}\cdot\vec{B} = AB\cos\theta$

$1 + 0 = 2 \times 1\cos\theta$

$\cos\theta = 1/2$

$\theta = 60°$

Q.36. If $\vec{A} = 2\hat{i} + 3\hat{j} + \alpha\hat{k}$ and $\vec{B} = \hat{i} - 2\hat{j} + 4\hat{k}$, find α if $\vec{A}$ is perpendicular to $\vec{B}$.

Sol. $\vec{A} = 2\hat{i} + 3\hat{j} + \alpha\hat{k}$

$\vec{B} = \hat{i} - 2\hat{j} + 4\hat{k}$

Since $\vec{A}$ is perpendicular to $\vec{B}$,

$\vec{A}\cdot\vec{B} = 0$

$(2)(1) + (3)(-2) + (\alpha)(4) = 0$

$2 - 6 + 4\alpha = 0$

$-4 + 4\alpha = 0$

$4\alpha = 4$

$\alpha = 1$

Q.37. If velocity $\vec{V} = \hat{i} - 2\hat{j} + \hat{k}$ and acceleration $\vec{a} = 2\hat{i} + \hat{j} + 2\hat{k}$, find the component of velocity along acceleration.

Sol. $\vec{V} = \hat{i} - 2\hat{j} + \hat{k}$

$\vec{a} = 2i + \hat{j} + 2\hat{k}$

$|\vec{a}| = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$

Component of $\vec{V}$ along $\vec{a} = (\vec{v}\cdot\vec{a})/a$

$= (\hat{i} - 2\hat{j} + \hat{k})\cdot(2\hat{i} + \hat{j} + 2\hat{k})/3$

$= (2 - 2 + 2)/3 = 2/3$

Q.38. Find the projection of $\vec{A} = 2\hat{i} - \hat{j} + \hat{k}$ along the vector $\vec{B} = \hat{i} + \hat{j} + \hat{k}$

Sol. $\vec{A}\cdot\vec{B} = AB\cos\theta$

$A\cos\theta = \vec{A}\cdot\vec{B}/B$

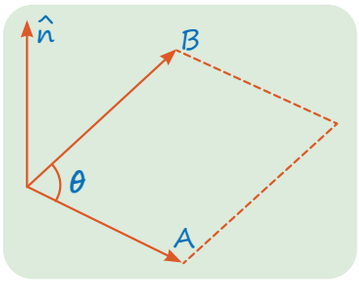
$= ((2)(1) + (-1)(1) + (1)(1))/\sqrt{3}$

$= 2/\sqrt{3}$

7 Cross-Product : [Vector Product]

Perpendicular component ko chipka do, ||r component ko ignore kro

- $\vec{A}\times\vec{B} = AB\sin\theta\,\hat{n}$



- $\hat{n}$ is direction of $\vec{A}\times\vec{B}$ which is perpendicular to $\vec{A}$ & $\vec{B}$.
- $(\vec{A}\times\vec{B})\cdot\vec{A} = 0 \quad (\vec{A}\times\vec{B})\cdot\vec{B} = 0$
- $B\sin\theta = \frac{\vec{A}\times\vec{B}}{A}$ = component of B perpendicular to A
- $\vec{R} = \vec{A}\times\vec{B}$

Place your finger of right hand along $\vec{A}$ and slap $\vec{B}$ then thumb will represent $\vec{R}$.

$\hat{i} \times \hat{i} = 0 = \hat{j} \times \hat{j} = \hat{k} \times \hat{k}$

$\hat{i} \times \hat{j} = \hat{k}$ $\quad$ $\hat{j} \times \hat{i} = -\hat{k}$

$\hat{j} \times \hat{k} = \hat{i}$ $\quad$ $\hat{k} \times \hat{j} = -\hat{i}$

$\hat{k} \times \hat{i} = \hat{j}$ $\quad$ $\hat{i} \times \hat{k} = -\hat{j}$

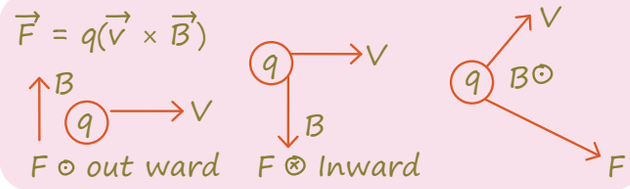
✦ $\vec{A} \times \vec{B} = \begin{vmatrix} i & j & k \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$

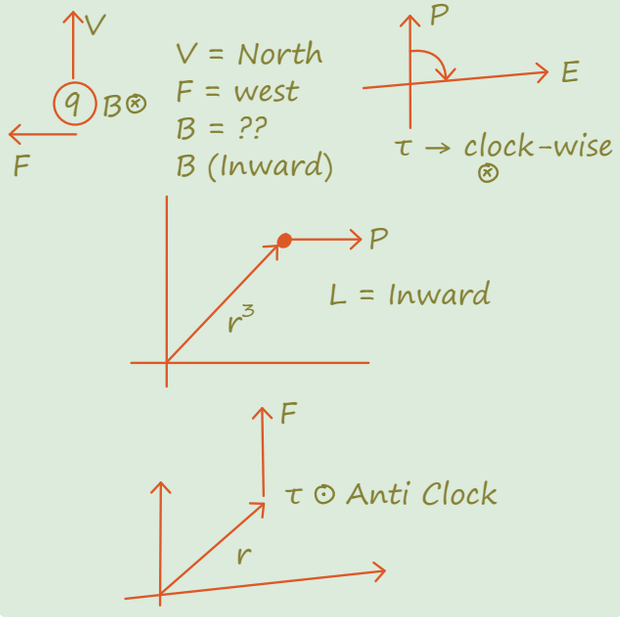
$= i(A_yB_z - A_zB_y) - J(A_xB_z - A_zB_x) + K(A_xB_y - A_yB_x)$

✦ $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ but $\vec{A} \times \vec{B} \neq \vec{B} \times \vec{A}$

Direction of cross product

$\vec{F} = q(\vec{v} \times \vec{B})$





- Unit vector does not have any unit only have direction and magnitude one.
- Minimum no. of vectors whose resultant can be zero is '2'.
- Minimum no of unequal vectors whose resultant can be zero is 3.
- The resultant of 3 Non- coplanar vectors can't be zero.
- Minimum no of Non-coplanar, vectors whose resultant can be zero is 4.
- Angle between $(\vec{A} \times \vec{B})$ and $(\vec{A} + \vec{B})$ is 90°
- Division of vector with vector is not possible
- Division of magnitude of vector is possible
- Vector can be divided by scalar.
- If vector multiplied by positive scalar then magnitude change and direction remains same.
- If vector multiplied by negative scalar then magnitude change and direction becomes opposite.

MR* BOX

Cross Product ho ya Dot Product use to physics me hi karna hai, to dekh lo kaha use hoga.

Dot Product	Cross Product
Work: $W = \vec{F} \cdot \vec{S}$	$\vec{\tau} = \vec{r} \times \vec{F}$
Power: $P = \vec{F} \cdot \vec{V}$	$\vec{L} = \vec{r} \times \vec{p}$
Potential Energy: $U = \vec{P} \cdot \vec{E}$	$\vec{F} = q(\vec{v} \times \vec{B})$
Flux: $\Phi = \vec{E} \cdot \vec{A}$	$\vec{v} = \vec{\omega} \times \vec{r}$
Gauss Law: $\phi = \vec{E} \cdot d\vec{A} = q_{in}/\varepsilon_0$	$\vec{\tau} = \vec{P} \times \vec{E}$
$\Delta V = \vec{E} \cdot \overrightarrow{\Delta l}$	$\vec{F} = \vec{l\ell} \times \vec{B}$
	$\overrightarrow{dB} = \frac{\mu_0}{4\pi} \frac{\overrightarrow{|dl| \times \vec{r}}}{r^3}$

MR Scam

Which of the following operation is possible:

(i) $(\vec{A} \cdot \vec{B}) \cdot \vec{C}$ → Not possible
Reason: $\vec{A} \cdot \vec{B}$ is a scalar, so scalar ka dot product nahi hota.

(ii) $(\vec{A} \cdot \vec{B})/\vec{C}$ → Not possible
Reason: Scalar / Vector is not defined.

(iii) $\vec{A}/(\vec{B} \cdot \vec{C})$ → Possible
Reason: $\vec{B} \cdot \vec{C}$ is scalar, Vector / Scalar is possible.

(iv) $\vec{A} \cdot B$ → Not possible.

(v) $(\vec{A} \times \vec{B}) \cdot \vec{C}$ → Possible
This is Scalar Triple Product.
Physics me bhi ye aata hai:
Motional EMF: $emf = (\vec{V} \times \vec{B}) \cdot \vec{\ell}$

Scalar triple Product

$R = (\vec{A} \times \vec{B}) \cdot \vec{C}$ Result R will be scalar and R will be zero if any of these two vector becomes parallel.

Area of parallelogram

$$\text{Area} = |\vec{A} \times \vec{B}|$$

Area of triangle:

$$\text{Area} = \frac{1}{2}|\vec{A} \times \vec{B}|$$

The condition for coplanarity is

$$\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$$

Q.39. Force acting on object $\vec{F} = 5\hat{i} + 3\hat{j} - 7\hat{k}$ position vector $\vec{r} = 2i + 2j - k$ then find torque ?? (NEET 2022)

Sol. $\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & -1 \\ 5 & 3 & -7 \end{vmatrix}$

$\hat{i}(-14 - (-3) - \hat{j}(-14 - (-5)) + \hat{k}(6-10)$

$-11\hat{i} + 9\hat{j} - 4\hat{k}$

Q.40. If $|\vec{A} \times \vec{B}| = \sqrt{3}\,\vec{A} \cdot \vec{B}$ then angle between $\vec{A}$ and $\vec{B}$ is?

Sol. $AB \sin\theta = \sqrt{3}\, AB \cos\theta$

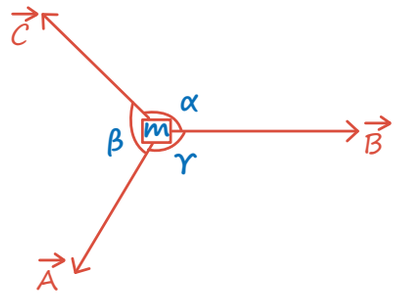
$\tan\theta = \sqrt{3} \Rightarrow \theta = 60°$

8 Lami's Theorem (Laws of motion) for 3 Vectors

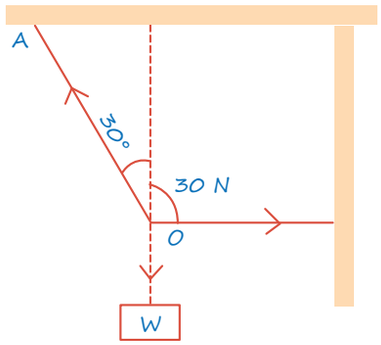
When,

$$\frac{A}{\sin\alpha} = \frac{B}{\sin\beta} = \frac{C}{\sin\gamma}$$

Then, $\vec{A} + \vec{B} + \vec{C} = 0$



Q.41. As shown in figure tension in the horizontal cord is 30 N. The weight W and tension in the string OA in Newton are



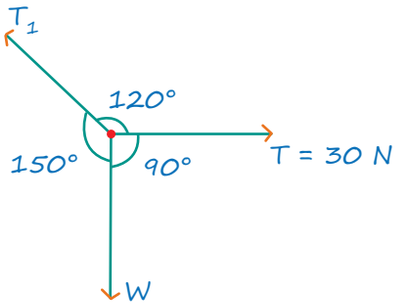
(a) $30\sqrt{3}$, 30 (b) $30\sqrt{3}$, 60
(c) $60\sqrt{3}$, 30 (d) None of these

Sol. $\frac{30}{\sin 150°} = \frac{T_1}{\sin 90°}$

$\frac{T_1}{1} = \frac{30}{1/2}$, $T_1 = 60$ N

$\frac{W}{\sin 120°} = \frac{30}{\sin 150°}$

$\frac{W}{\sqrt{3}/2} = \frac{30}{1/2} = 30\sqrt{3}$



MR*

अपनी पढाई छोड़ जो तेरे पीछे चला आयेगा।
वो खुद का ना हो सका, तेरा क्या हो जायेगा।।